

CAR RENTAL REQUEST AUTOMATION IN SERVICENOW

Phase 2: Requirement Analysis

Turns the idea into specific, checkable requirements that can be traced to the build.

Item	Details
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Program	SkillWallet, ServiceNow Global Certification
Platform	ServiceNow (Service Catalog, Flow Designer, Business Rules)
Document	Phase 2 of 5

In this document

- 2.1 Requirements Gathering Approach
- 2.2 Functional Requirements
- 2.3 Business Rules and Policies
- 2.4 Non-Functional Requirements
- 2.5 Data Requirements
- 2.6 Roles and Access Requirements
- 2.7 System and Technical Requirements
- 2.8 User Stories
- 2.9 Use Cases and Scenarios
- 2.10 Requirements Traceability
- 2.11 Assumptions, Constraints and Dependencies

2.1 Requirements Gathering Approach

Requirements were derived from three sources: the project brief and Phase 1 objectives on SkillWallet, the stakeholder needs identified in Phase 1, and the capabilities and constraints of the ServiceNow Service Catalog and Flow Designer. Each requirement below has an identifier so that it can be traced to a design element in Phase 3 and a build step in Phase 5.

2.2 Functional Requirements

ID	Requirement	Acceptance criteria	Realized by
FR-1	Users can submit a car rental request through a catalog item	The item is visible in the Service Catalog and can be ordered	Catalog item "Car Rental Request"
FR-2	The form captures requestor, date, pickup, drop, duration, car type, and reason	All seven fields appear on the form and their values are saved with the request	Seven catalog variables
FR-3	Car type is chosen from a fixed list	Only Mini, SUV, and Sedan can be selected	Select Box variable

ID	Requirement	Acceptance criteria	Realized by
FR-4	Submission automatically starts a workflow	Ordering the item starts an execution of the flow	Flow Designer, Service Catalog trigger
FR-5	The request is sent to an approver	An approval record is created and the request waits for a decision	Ask For Approval action
FR-6	If approved, a task is created for the Transport team	An sc_task exists with assignment group Transport Team	Create Catalog Task
FR-7	The requester is notified by email	An email with the agreed subject is sent to the Requestor	Send Email action
FR-8	An incident is raised when a car is unavailable	An incident referencing the request number is inserted	Business Rule on Requested Item
FR-9	Request status is trackable	REQ, RITM, and task records show the current state	Catalog records
FR-10	The configuration is portable	The changes can be exported and committed on another instance	Update Set

2.3 Business Rules and Policies

ID	Rule
BR-1	No fulfilment work starts until the request is approved
BR-2	Every approved request is assigned to the Transport Team, never to an individual
BR-3	Every approval decision is recorded on the request in the Approval history journal
BR-4	The requester is told the outcome by email
BR-5	A car that cannot be provided is escalated by raising an incident that quotes the request number

2.4 Non-Functional Requirements

Category	Requirement	How it is met
Usability	Clean form with clear labels, help text, and tooltips	Short form, controlled Car Type list, portal layout
Traceability	Every decision and interaction can be reviewed later	Approval history journal; REQ, RITM, task records
Security	Only the designated approver can approve	Approval rule names the approver
Maintainability	Admins can change the logic without heavy coding	Visual flow in Flow Designer
Portability	The solution can move between instances	All changes captured in an Update Set

Category	Requirement	How it is met
Reliability	Approved requests are never left unassigned	Task creation is part of the same flow, not a manual step

2.5 Data Requirements

#	Label	Type	Name	Notes
1	Requestor	Reference	requestor	User the request is for; used to address the email
2	Requested Date	Date	requested_date	Date the car is needed
3	Pickup Location	Single Line Text	pickup_location	
4	Drop Location	Single Line Text	drop_location	
5	Duration (Hours)	Numeric Scale	duration_hrs	Options 0 to 5
6	Car Type	Select Box	car_type	Mini, SUV, Sedan
7	Reason	Multi Line Text	reason	Business justification

Data validation expectations. Car Type must be one of the three listed values. Requested Date should be a valid date. Duration is limited to the scale offered. Stronger validation, such as rejecting past dates, is listed as future work.

2.6 Roles and Access Requirements

Role	What they must be able to do
Requester	Open the catalog, order the item, and view their own request
Approver	See the approval request and approve or reject it
Transport Team member	See and work the catalog tasks assigned to the Transport Team group
ServiceNow admin	Create and maintain the catalog item, variables, flow, business rule, and Update Set

2.7 System and Technical Requirements

Requirement	Detail
ServiceNow instance	A configured instance with administrative access
Service Catalog module	Access to the Service Catalog and related features
Workflow engine	Flow Designer (Workflow Studio) to automate the request process
Scripting	Business Rules (server-side JavaScript) for incident creation

Requirement	Detail
Skills applied	Batch updates (Update Sets), Service Catalog, Flow Designer and process automation, business rules and client scripting

2.8 User Stories

Story	Acceptance
As an employee , I want to fill in one form with my travel details, so that my request is captured correctly the first time.	All seven fields are on the form and saved with the request
As an approver , I want approval requests to arrive automatically, so that I can decide without being chased.	An approval record appears after the order is placed
As a Transport team member , I want an approved request to appear as a task for my group, so that I know exactly what to arrange.	A catalog task is assigned to the Transport Team
As an admin , I want the logic in a visual flow, so that I can change it without rebuilding the solution.	The flow can be opened and edited in Flow Designer

2.9 Use Cases and Scenarios

Main scenario: request approved and fulfilled

1. The requester opens Car Rental Request, completes the form, and clicks Order Now.
2. The system creates a REQ and a RITM and starts the flow.
3. The approver approves the request.
4. The flow creates a catalog task for the Transport Team.
5. When the task is completed, the requester receives the approval email.

Alternate scenarios

Scenario	Behaviour in the current build
Request is rejected	The Approval State is not Approved, so the If branch does not run: no task and no email. A rejection notice is listed as future work.
Car is unavailable	The business rule inserts an incident quoting the request number and the requested date.

2.10 Requirements Traceability

Objective	Requirements	Where it is built
1. Standard catalog item	FR-1, FR-2, FR-3	Phase 5, Section 5.3
2. Automated approval	FR-4, FR-5, BR-1, BR-3	Phase 5, Section 5.4 (Activities 1 to 3)
3. Automatic task assignment	FR-6, BR-2	Phase 5, Section 5.4 (Activity 4)

Objective	Requirements	Where it is built
4. Notification and escalation	FR-7, FR-8, BR-4, BR-5	Phase 5, Sections 5.4 (Activities 5, 6) and 5.5
5. Centralized tracking	FR-9	Catalog records (Phase 3, Section 3.4)
6. Better user experience	FR-1 to FR-3, FR-7	Phase 3, Section 3.5; Phase 5, Section 5.8
Portability	FR-10	Phase 5, Section 5.9

2.11 Assumptions, Constraints and Dependencies

- One approver is configured in the approval rule ("Anyone approves").
- One assignment group, the Transport Team, receives every approved task.
- The email goes to the user selected in the Requestor variable.
- The project is built and demonstrated on a ServiceNow developer instance.
- **Dependency:** the flow needs the catalog item and its variables to exist first, because it reads them as data pills.